SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: <u>S101 Data Analysis with SAS / SPSS / R Practical</u>

PRACTICAL - 12 (M2)

AIM: Generating correlation matrices using cor() (R).

Input:

```
# S101 RAJDEEPP .M. PARAB
# Data Analysis with SAS / SPSS / R Practical
# PRACTICAL 12 – Correlation Matrix in R
# Dataset: water_potability.csv
# Load libraries
library(dplyr)

# 1. Load dataset
df <- read.csv("water_potability.csv")

# 2. Keep only numeric columns
num_df <- df %>% select_if(is.numeric)

# 3. Remove missing values
num_df <- na.omit(num_df)


# 4. Compute correlation matrix
cor_matrix <- cor(num_df)

# Print full matrix
print(cor_matrix)

# Rounded clean matrix view
round(cor_matrix, 2)

# 5. Install and load corrplot (for graph)
install.packages("corrplot")
library(corrplot)

# 6. Draw correlation heatmap graph
corrplot(cor_matrix,
      method = "color",   # color heatmap
      tl.cex = 0.7,       # label size
      number.cex = 0.7)   # number size
```

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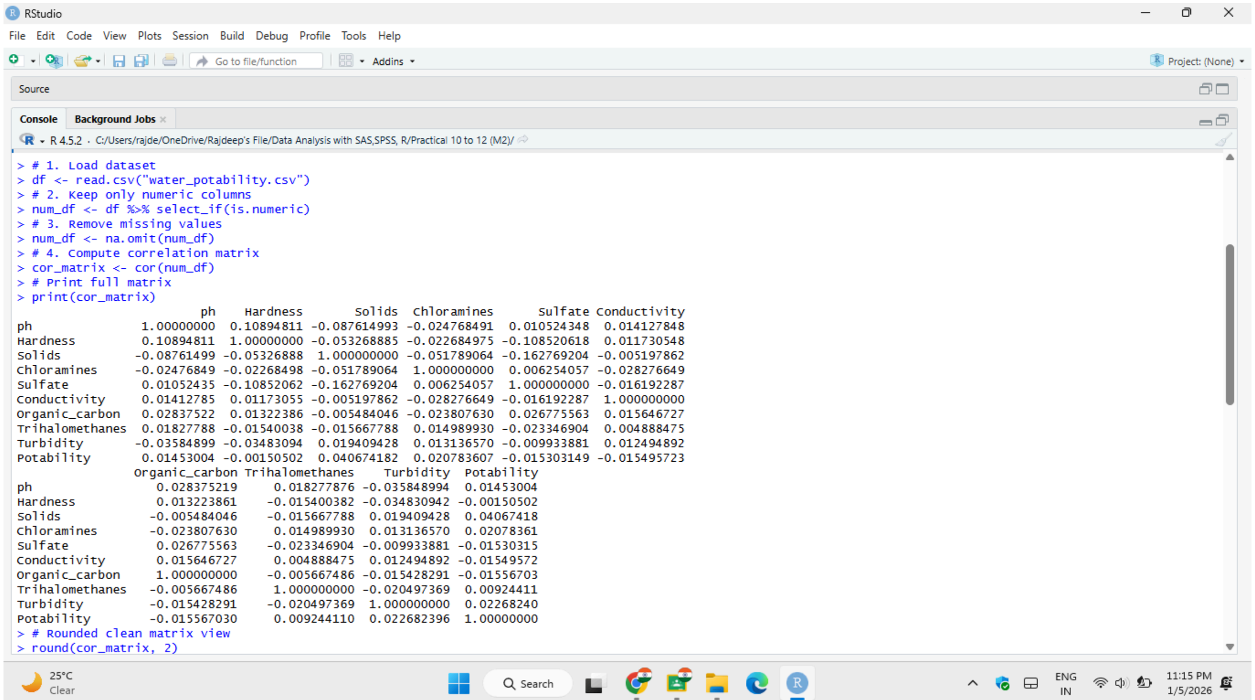
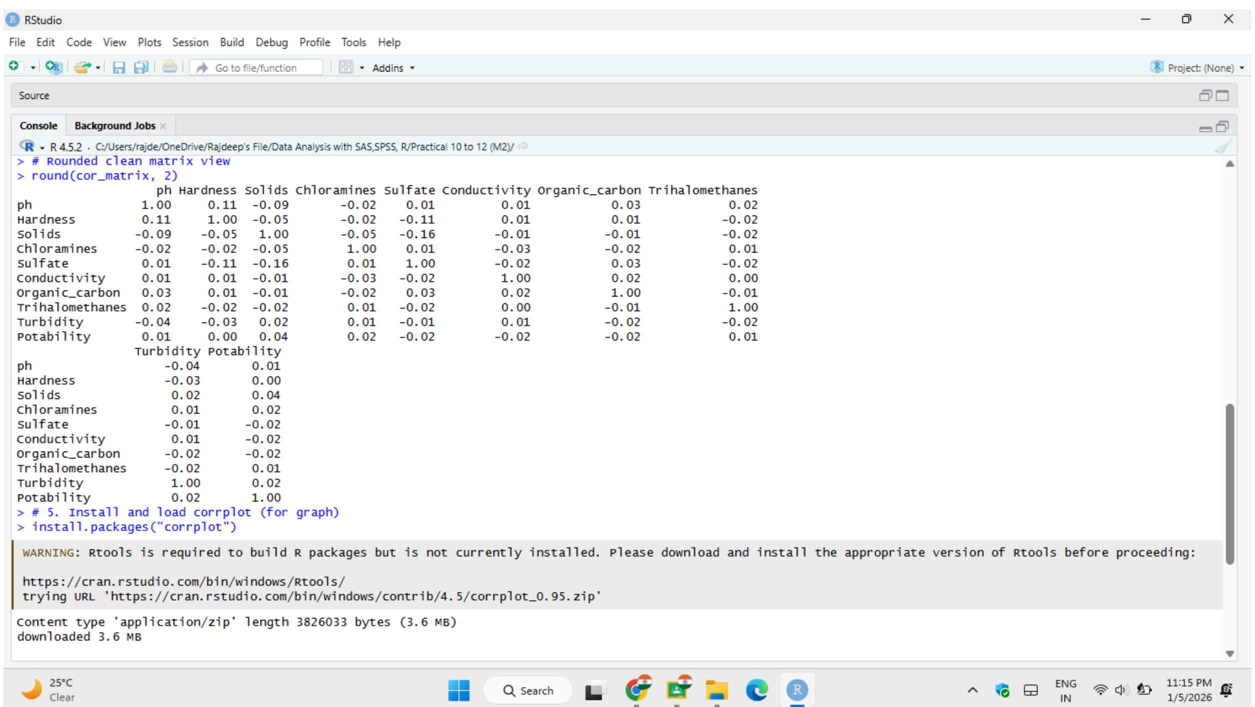
Output:



```
> # 1. Load dataset
> df <- read.csv("water_potability.csv")
> # 2. Keep only numeric columns
> num_df <- df %>% select_if(is.numeric)
> # 3. Remove missing values
> num_df <- na.omit(num_df)
> # 4. Compute correlation matrix
> cor_matrix <- cor(num_df)
> # Print full matrix
> print(cor_matrix)
                        ph    Hardness       Solids  Chloramines      Sulfate Conductivity
ph              1.00000000  0.10894811 -0.087614993 -0.024768491  0.010524348  0.014127848
Hardness        0.10894811  1.00000000 -0.053268885 -0.022684975 -0.108520618  0.011730548
Solids         -0.08761499 -0.05326888  1.000000000 -0.051789064 -0.162769204 -0.005197862
Chloramines    -0.02476849 -0.02268498 -0.051789064  1.000000000  0.006254057 -0.028276649
Sulfate         0.01052435 -0.10852062 -0.162769204  0.006254057  1.000000000 -0.016192287
Conductivity    0.01412785  0.01173055 -0.005197862 -0.028276649 -0.016192287  1.000000000
Organic_carbon  0.02837522  0.01322386 -0.005484046 -0.023807630  0.026775563  0.015646727
Trihalomethanes 0.01827788 -0.01540038 -0.015667788  0.014989930 -0.023346904  0.004888475
Turbidity      -0.03584899 -0.03483094  0.019409428  0.013136570 -0.009933881  0.012494892
Potability      0.01453004 -0.00150502  0.040674182  0.020783607 -0.015303149 -0.015495723
               Organic_carbon Trihalomethanes    Turbidity  Potability
ph                0.028375219     0.018277876 -0.035848994  0.01453004
Hardness          0.013223861    -0.015400382 -0.034830942 -0.00150502
Solids           -0.005484046    -0.015667788  0.019409428  0.04067418
Chloramines      -0.023807630     0.014989930  0.013136570  0.02078361
Sulfate           0.026775563    -0.023346904 -0.009933881 -0.01530315
Conductivity      0.015646727     0.004888475  0.012494892 -0.01549572
Organic_carbon    1.000000000    -0.005667486 -0.015428291 -0.01556703
Trihalomethanes  -0.005667486     1.000000000 -0.020497369  0.00924411
Turbidity        -0.015428291    -0.020497369  1.000000000  0.02268240
Potability       -0.015567030     0.009244110  0.022682396  1.00000000
> # Rounded clean matrix view
> round(cor_matrix, 2)
```



```
> # Rounded clean matrix view
> round(cor_matrix, 2)
                   ph Hardness Solids Chloramines Sulfate Conductivity Organic_carbon Trihalomethanes
ph               1.00     0.11  -0.09       -0.02    0.01         0.01           0.03            0.02
Hardness         0.11     1.00  -0.05       -0.02   -0.11         0.01           0.01           -0.02
Solids          -0.09    -0.05   1.00       -0.05   -0.16        -0.01          -0.01           -0.02
Chloramines     -0.02    -0.02  -0.05        1.00    0.01        -0.03          -0.02            0.01
Sulfate          0.01    -0.11  -0.16        0.01    1.00        -0.02           0.03           -0.02
Conductivity     0.01     0.01  -0.01       -0.03   -0.02         1.00           0.02            0.00
Organic_carbon   0.03     0.01  -0.01       -0.02    0.03         0.02           1.00           -0.01
Trihalomethanes  0.02    -0.02  -0.02        0.01   -0.02         0.00          -0.01            1.00
Turbidity       -0.04    -0.03   0.02        0.01   -0.01         0.01          -0.02           -0.02
Potability       0.01     0.00   0.04        0.02   -0.02        -0.02          -0.02            0.01
               Turbidity Potability
ph                 -0.04       0.01
Hardness           -0.03       0.00
Solids              0.02       0.04
Chloramines         0.01       0.02
Sulfate            -0.01      -0.02
Conductivity        0.01      -0.02
Organic_carbon     -0.02      -0.02
Trihalomethanes    -0.02       0.01
Turbidity           1.00       0.02
Potability          0.02       1.00
> # 5. Install and load corrplot (for graph)
> install.packages("corrplot")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/corrplot_0.95.zip'
Content type 'application/zip' length 3826033 bytes (3.6 MB)
downloaded 3.6 MB
```

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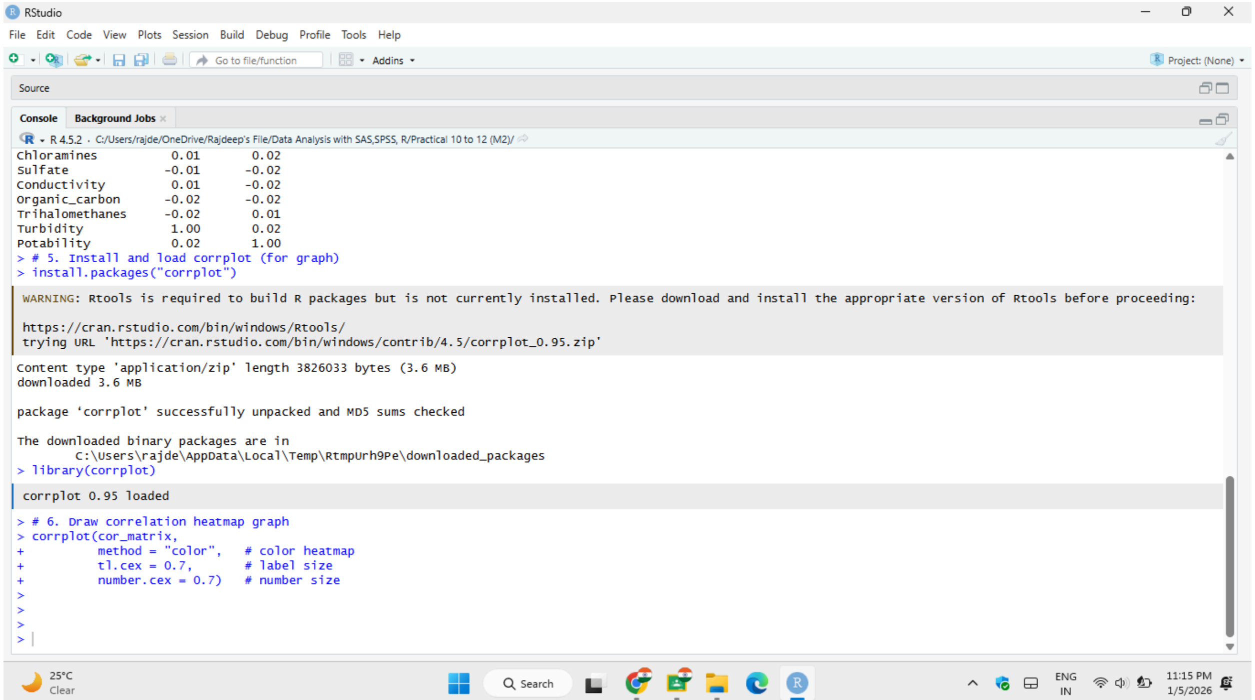

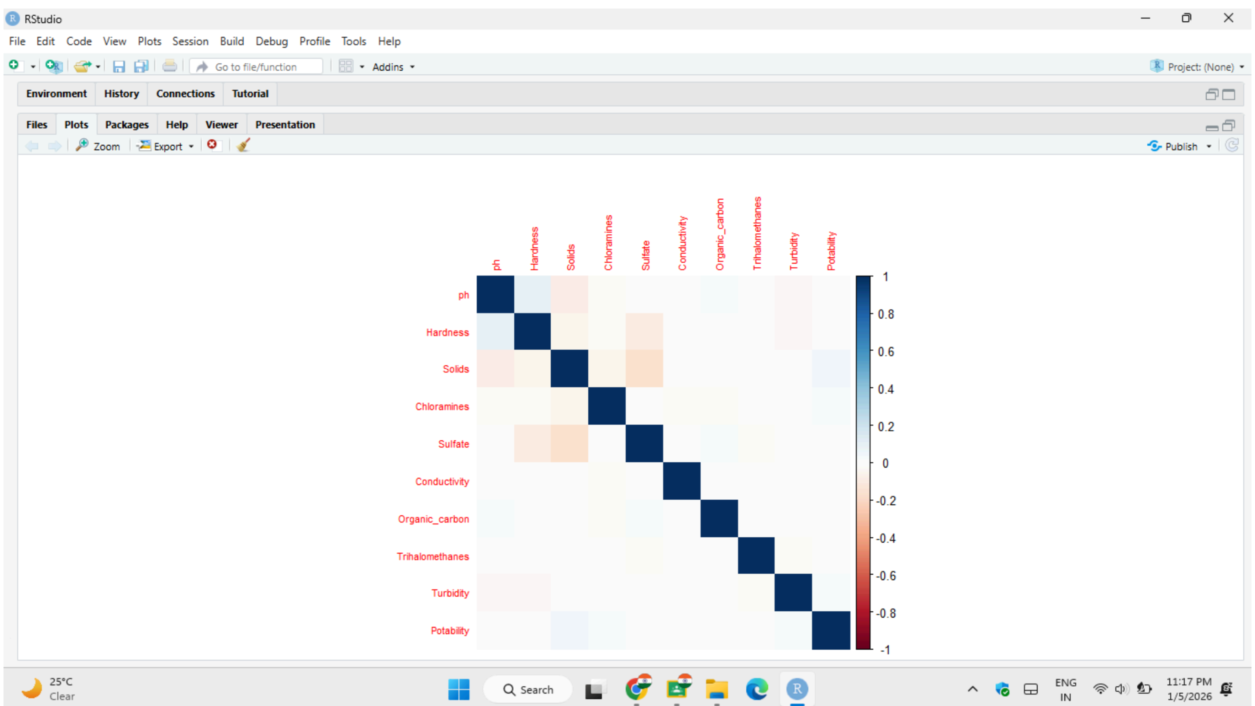

S101

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